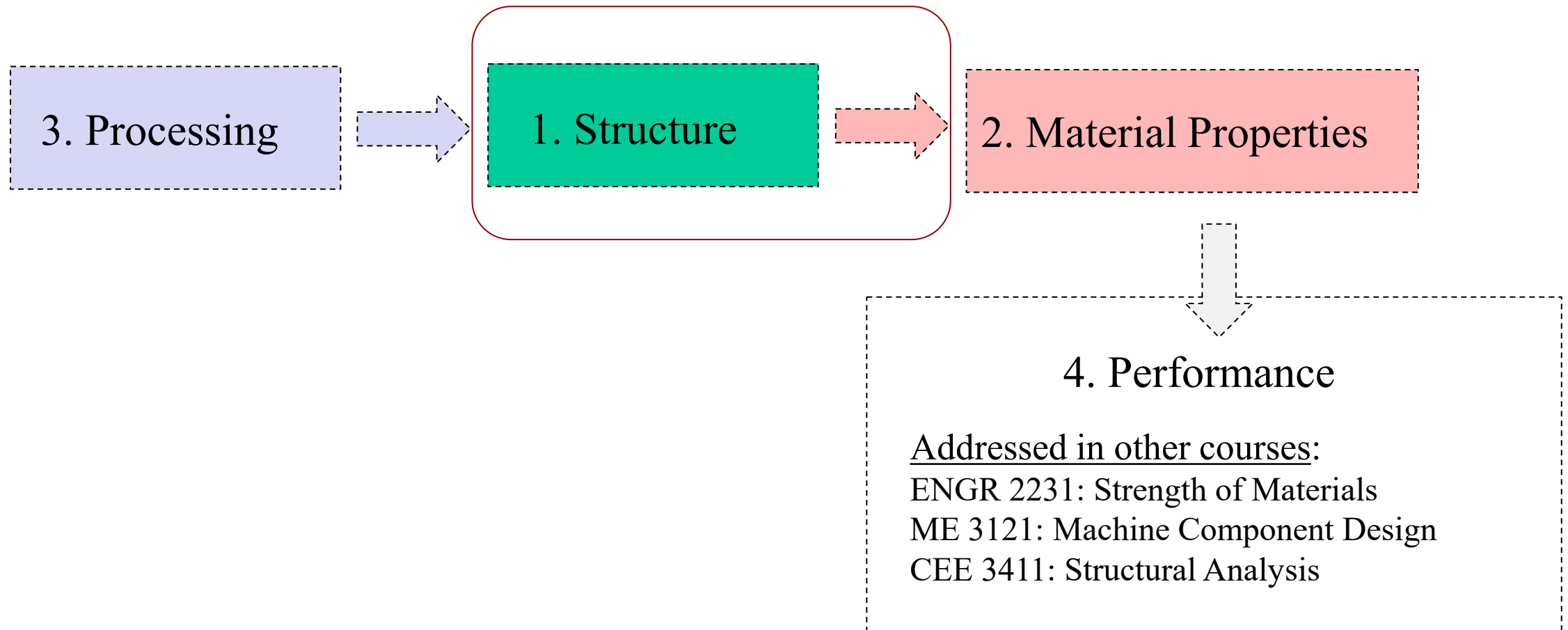


Material Science

Lecture 3: Imperfections

Today's Lecture

We are here!



Which one has higher tensile strength?



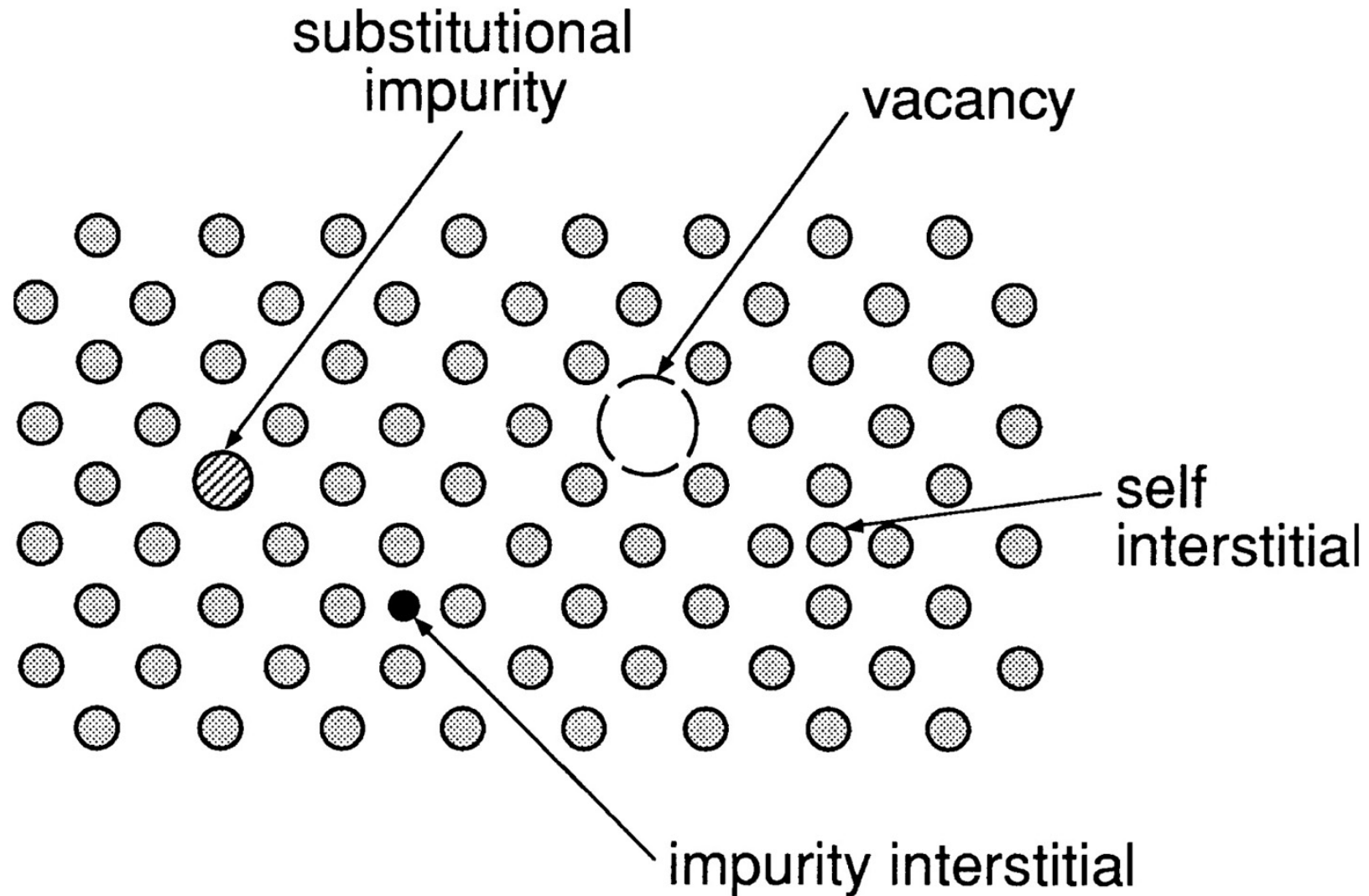
CH4 - Imperfections in Solids

There is no such thing as a perfect crystal.

- What are these imperfections?**
- Why are they important?**

Many of the important properties of materials are due to the presence of imperfections.

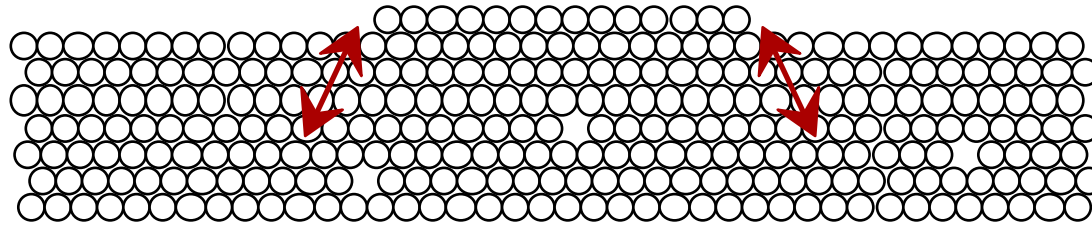
Point Defects



Observing Equilibrium Vacancy Conc.

Increasing temperature causes surface island of atoms to grow – Vacancies increase with temperature

Island grows/shrinks to maintain
equil. vacancy conc. in the bulk.



Why?

The equil. vacancy conc. increases via atom motion from the crystal to the surface, where they join the island.

Imperfections are not always bad!

Pure Al_2O_3



+ Fe

+ Co

Ruby



Sapphire



In a few weeks we'll show how foreign atoms make the host solid stronger!

Linear Defects (Dislocations)

DEF: one-dimensional defects around which atoms are misaligned

- **Edge dislocation:**
 - extra half-plane of atoms inserted in a crystal structure
 - $\mathbf{b}$ perpendicular ($\perp$) to dislocation line
- **Screw dislocation:**
 - spiral planar ramp resulting from shear deformation
 - $\mathbf{b}$ parallel ($\parallel$) to dislocation line

Burger's vector, $\mathbf{b}$: measure of lattice distortion

Imperfections in Solids

Edge Dislocation

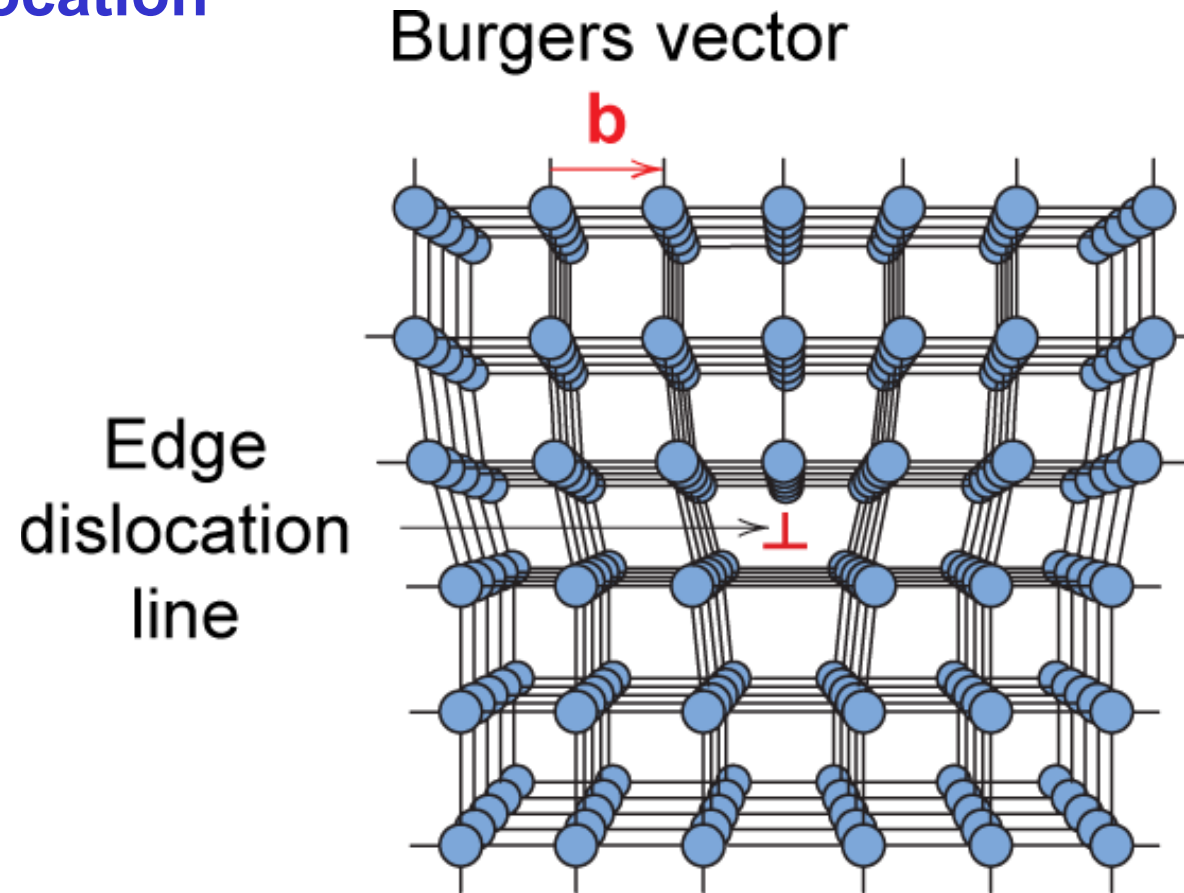
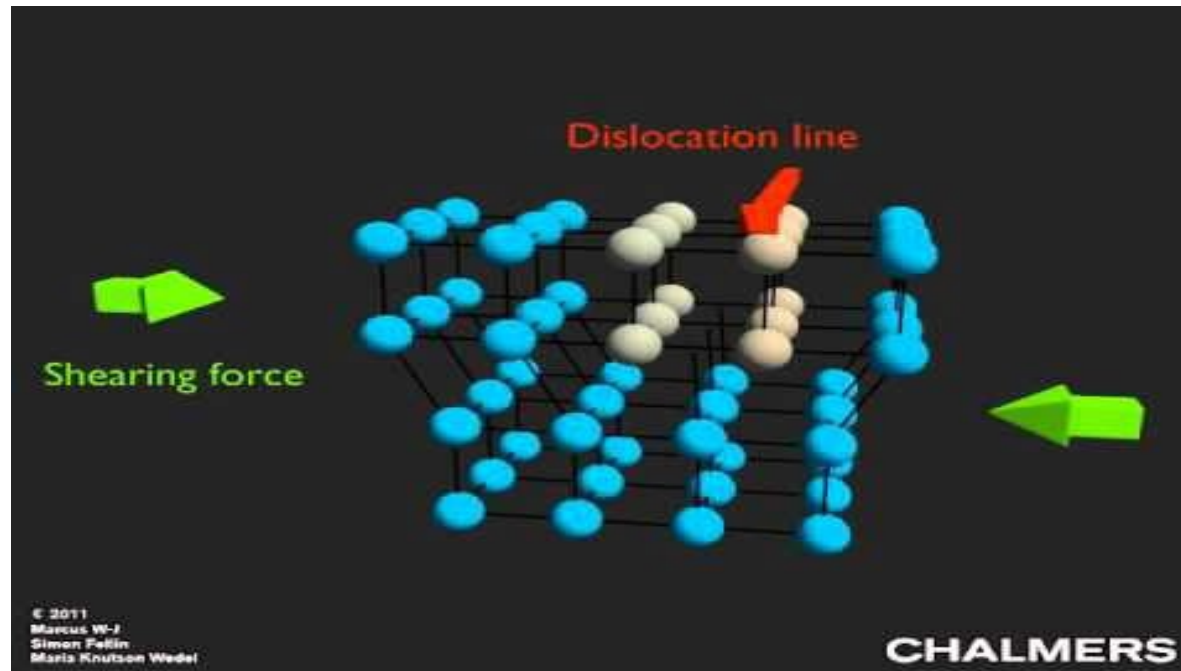


Fig. 5.8, Callister & Rethwisch 3e.

Motion of Edge Dislocation

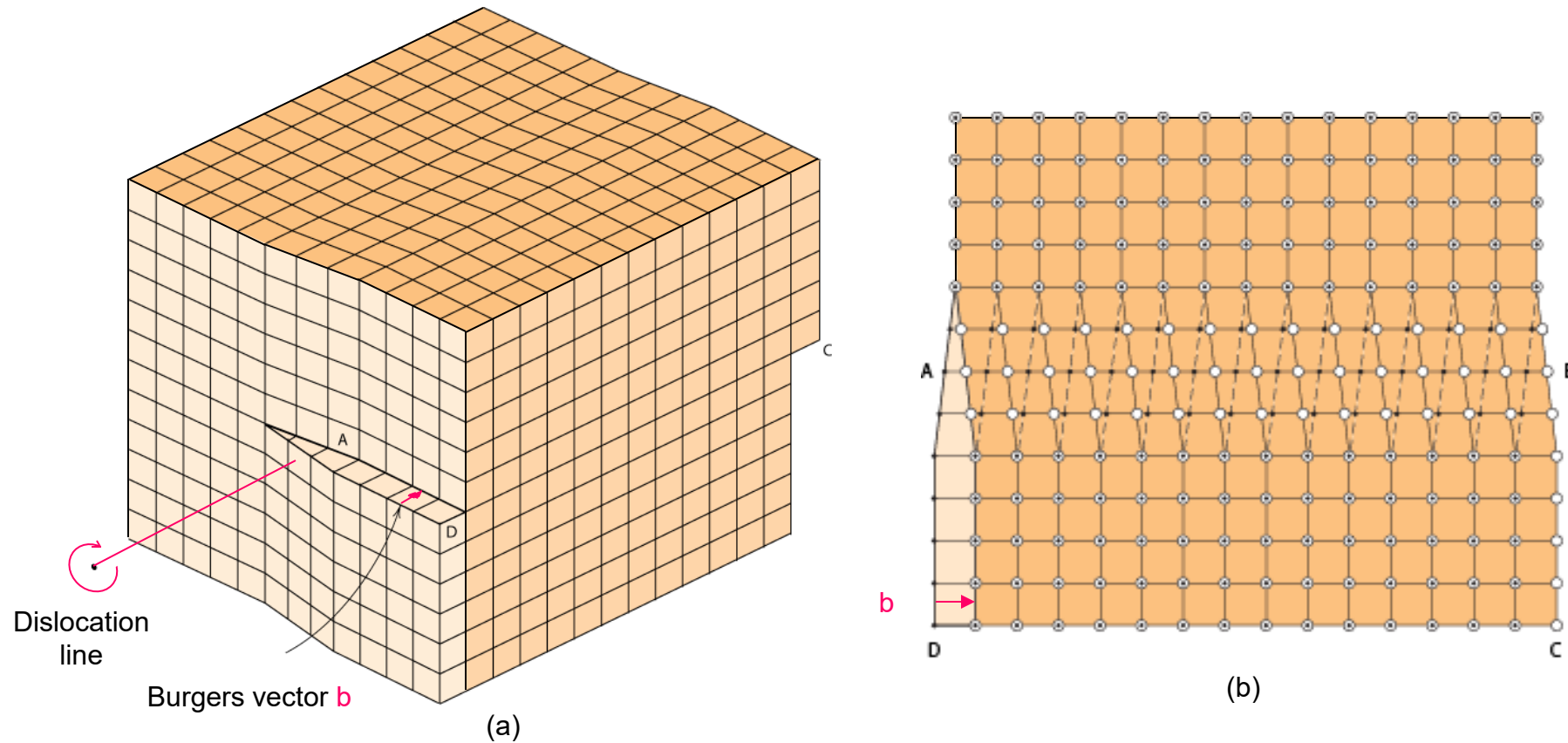
- Dislocation motion requires the successive bumping of a half plane of atoms (from left to right here).
- Bonds across the slipping planes are broken and remade in succession.

Atomic view of edge dislocation motion from left to right as a crystal is sheared.



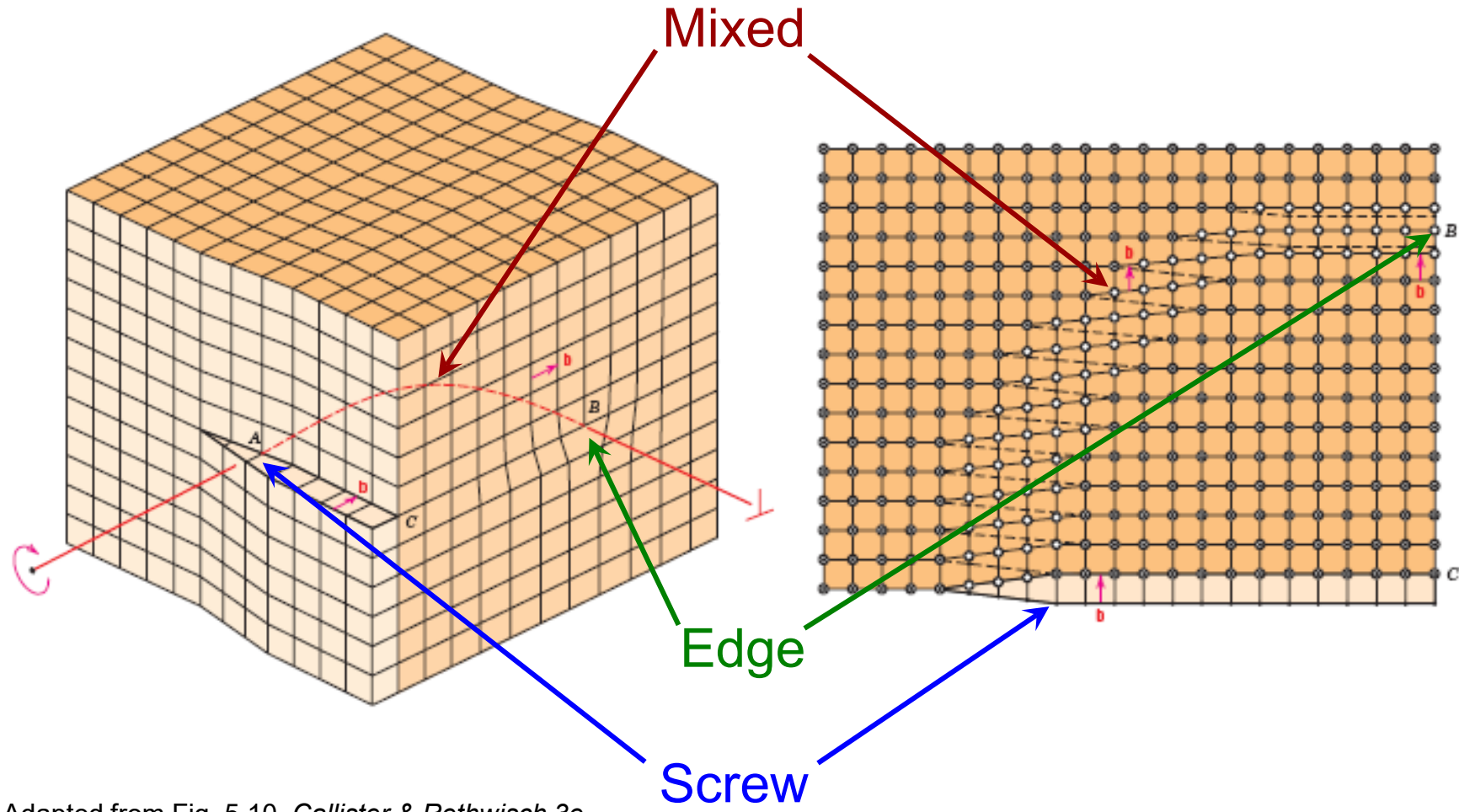
Imperfections in Solids

Screw Dislocation



Adapted from Fig. 5.9, *Callister & Rethwisch 3e*.

Edge, Screw, and Mixed Dislocations



Adapted from Fig. 5.10, Callister & Rethwisch 3e.

Imperfections in Solids

Dislocations are visible in electron micrographs

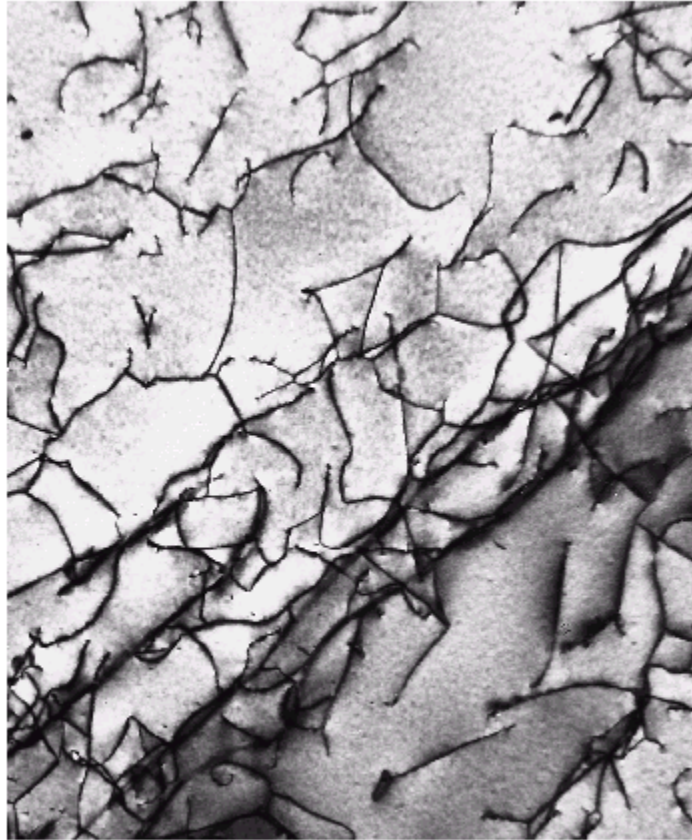


Fig. 5.11, *Callister & Rethwisch 3e.*

Crystalline imperfections can be used by engineers!

- **Dislocations allow crystals to deform rather than fracture (Properties)**
- **This makes the shaping of metals practical!**
- **Heat treatment or cold working can strengthen a material to prevent further dislocation motion (Processing)**

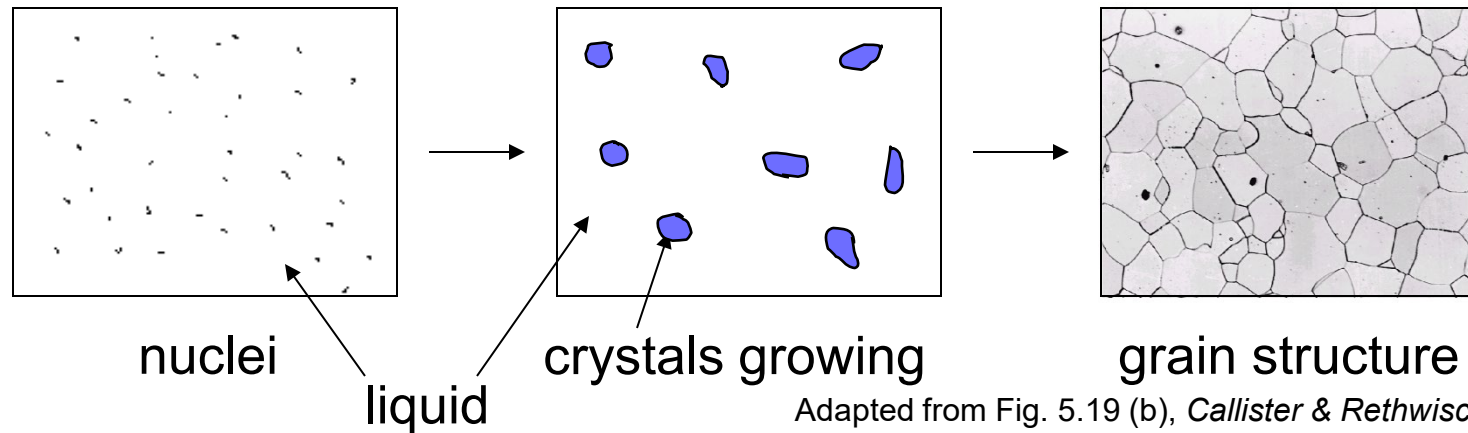
How is steel made?



<https://www.youtube.com/watch?v=efZsULWn1KY>

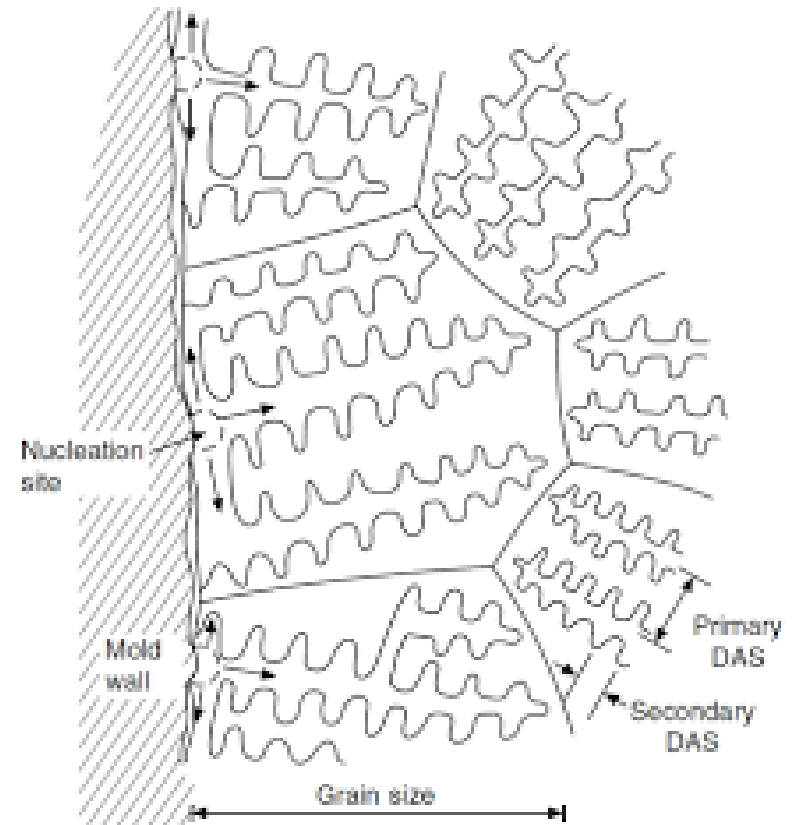
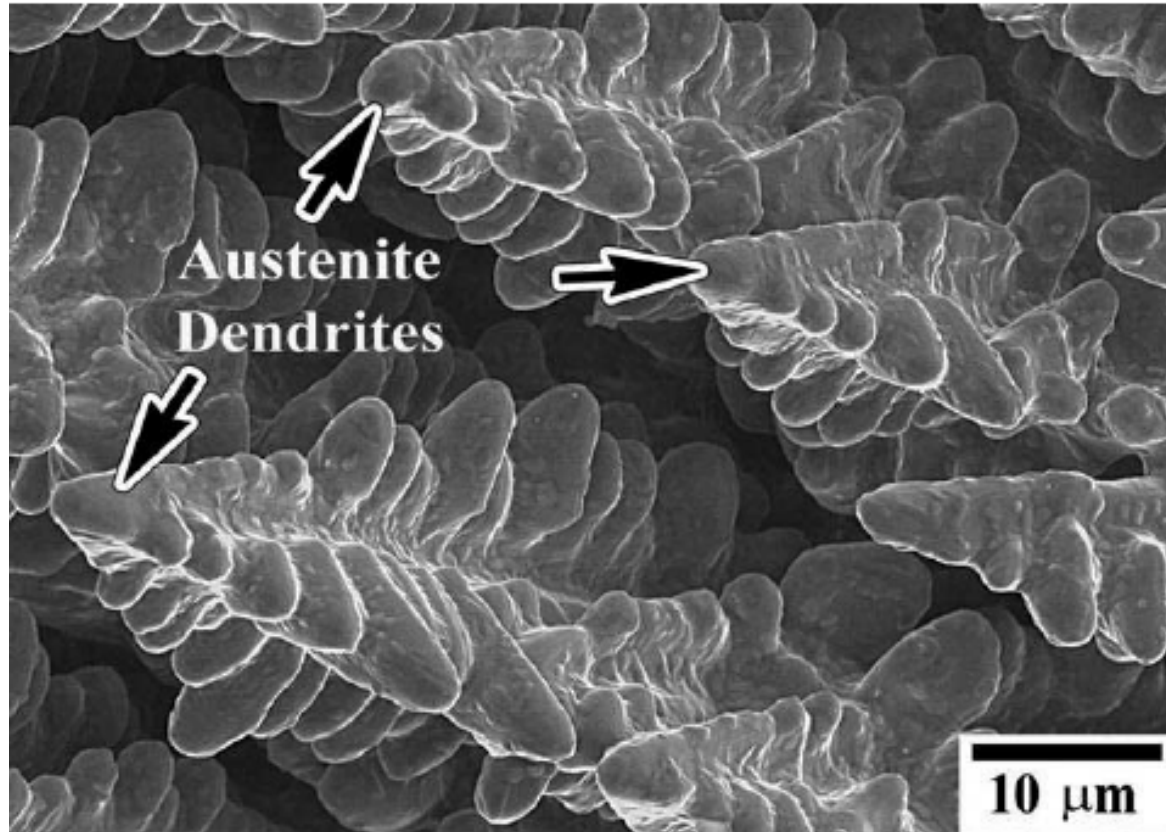
Interfacial (area) Defects

- **Solidification**- result of casting of molten material
 1. Nuclei form
 2. Nuclei grow to form crystals – grain structure



Crystals grow until they meet each other → Grain Boundary

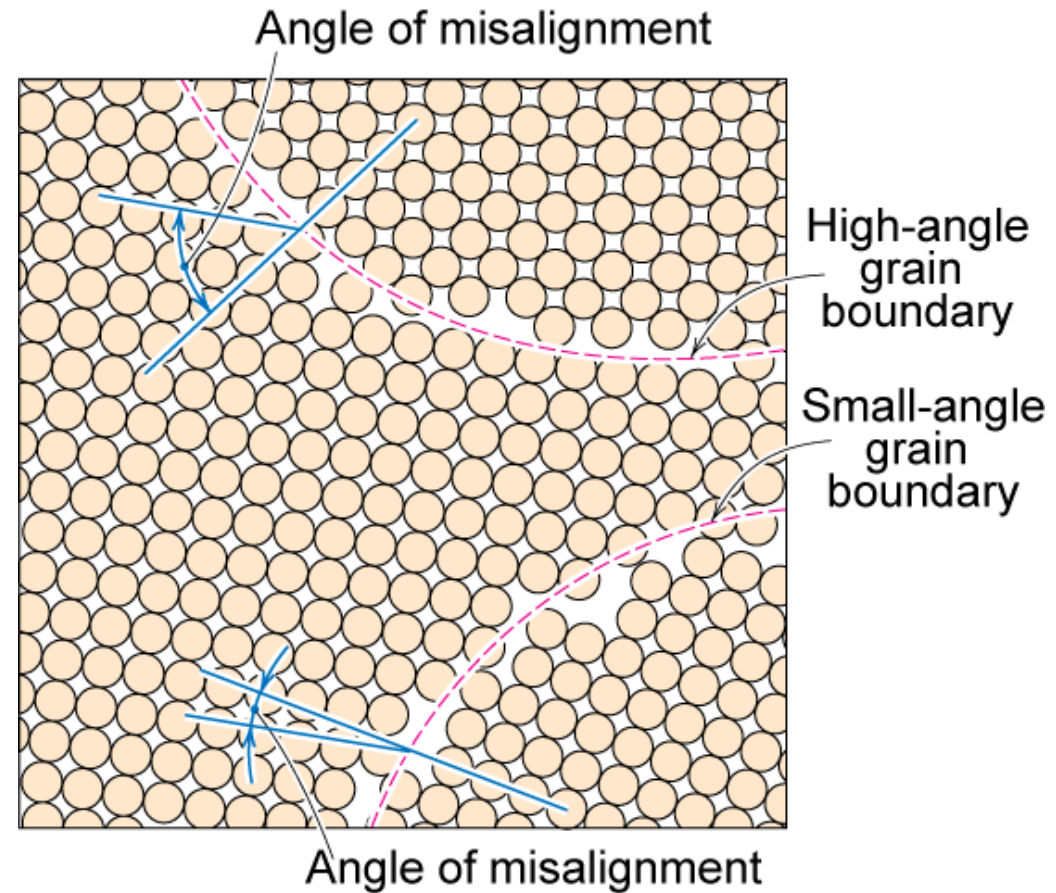
Dendrite Formation



Polycrystalline Materials

Grain Boundaries

- low density in grain boundaries
- high mobility and high chemical reactivity
- block dislocation motion



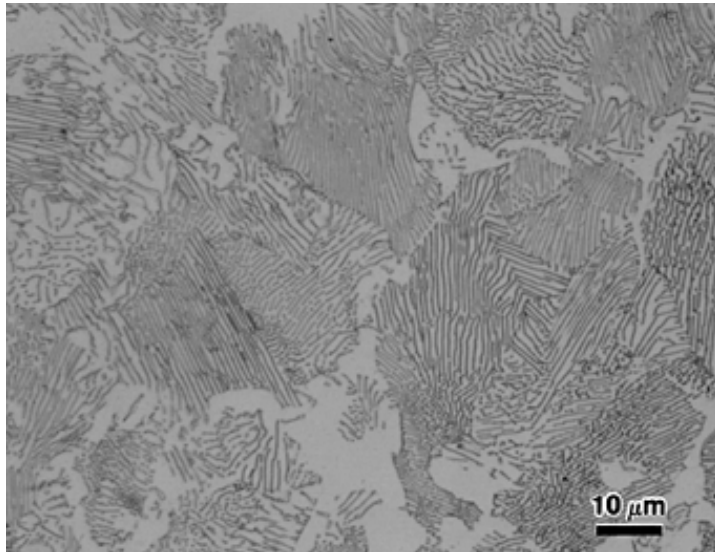
Adapted from Fig. 5.12,
Callister & Rethwisch 3e.

Microscopic Examination

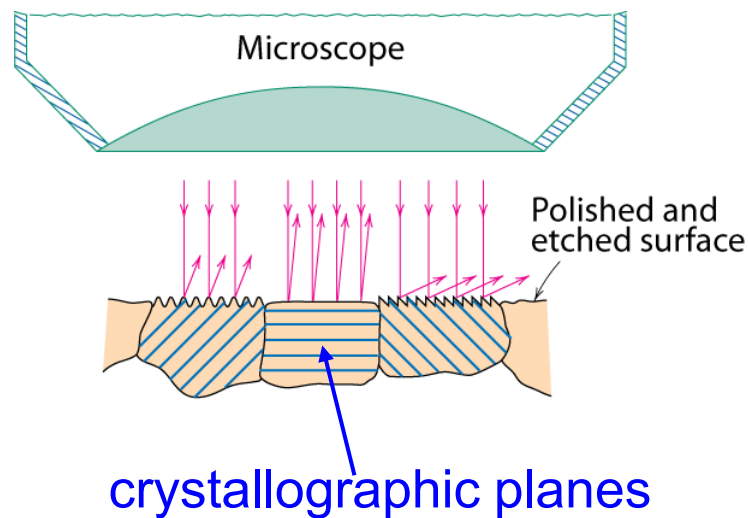
- **Crystallites (grains) and grain boundaries vary considerably in size and can be quite large.**
 - **ex: Large single crystal of quartz or diamond or Si**
- **Crystallites (grains) can be quite small (mm or less) – necessary to observe with a microscope.**

Optical Microscopy

- Useful up to 2000X magnification.
- Polishing removes surface features (e.g., scratches)
- Etching is a chemical process used to reveal microstructures
 - changes reflectance, depending on crystal orientation.



4140 Steel



← 0.75mm →



Micrograph of
brass (a Cu-Zn alloy)

Optical Microscopy

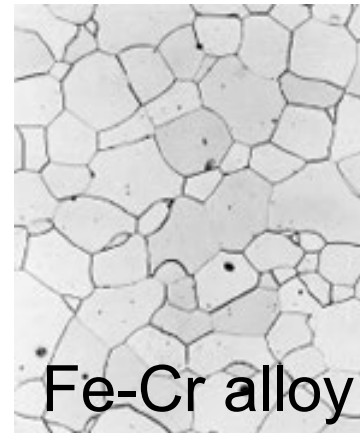
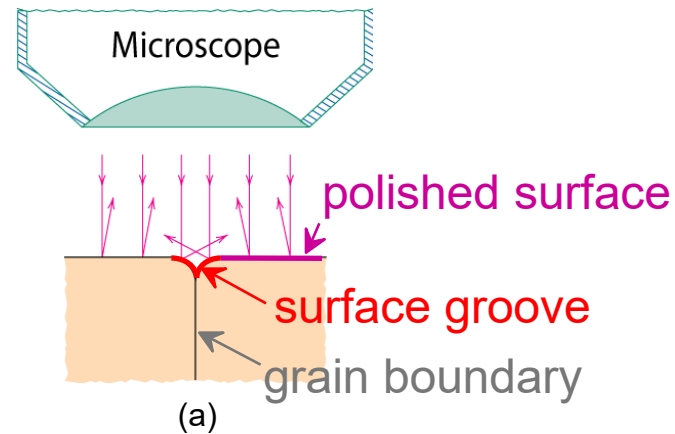
Grain boundaries...

- are imperfections
- are more susceptible to etching
- may be revealed as dark lines
- change in crystal orientation across boundary.

ASTM grain size number

$$N = 2^{n-1}$$

number of grains/in² at 100x magnification



Adapted from Fig. 5.19(a) and (b), *Callister & Rethwisch 3e*.

(Fig. 5.19(b) is courtesy of L.C. Smith and C. Brady, the National Bureau of Standards, Washington, DC [now the National Institute of Standards and Technology, Gaithersburg, MD].)

Practice Calculations

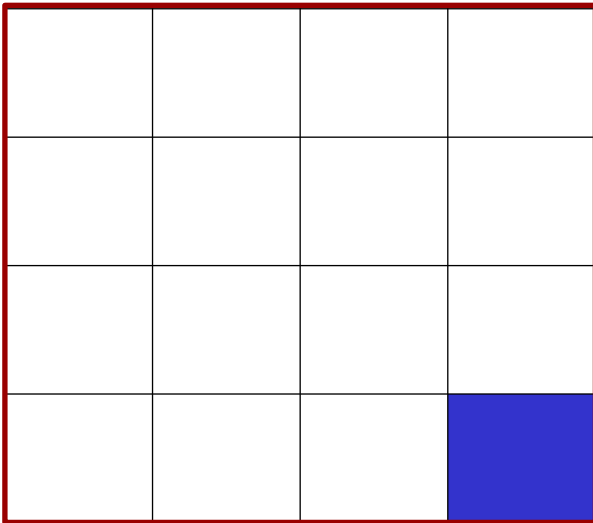
**Calculate the ASTM Grain Size Number for the following condition:
Approximately 4 grains seen in a 1"x1" area at 400X mag.**

Practice Calculations

**Calculate the ASTM Grain Size Number for the following condition:
Approximately 4 grains seen in a 1"x1" area at 400X mag.**

First, determine the # of grains at 100x mag. We have 2 options:

Option 1: Visual



Red square → 100x

Blue square → 400x

So... 100x has 16x the grains

Total grains at 100x (N) = 4 grains x 16

N = 64 grains

Practice Calculations

**Calculate the ASTM Grain Size Number for the following condition:
Approximately 4 grains seen in a 1"x1" area at 400X mag.**

First, determine the # of grains at 100x mag. We have 2 options:

Option 2: Calculation → Find N_{100}

$$N_{100} * 100^2 = N_{\text{mag}} * \text{mag}^2$$

$$N_{100} = N = \frac{4 \text{ grains} * 400^2}{100^2}$$

$$N = 4 \text{ grains} * 16$$

$$\underline{N = 64 \text{ grains}}$$

Known Values:

$$N_{\text{mag}} = 4$$

$$\text{mag} = 400$$

$$N = N_{100}$$

Practice Calculations

**Calculate the ASTM Grain Size Number for the following condition:
Approximately 4 grains seen in a 1"x1" area at 400X mag.**

From before: $N = 64$ Grains...

Now use ASTM Grain size equation to find n

$$N = 2^{n-1}$$

$$64 = 2^{n-1}$$

$$2^6 = 2^{n-1}$$

So... $n = 7$

Practice Calculations

**Calculate the ASTM Grain Size Number for the following condition:
Approximately 45 grains can be seen in a 1"x1" area at 100X mag.**

$N = 2^{n-1}$ How do you solve for n ? Take the natural log* of each side:

$$\log N = \log 2^{n-1}$$

$$\log N = (n - 1)\log 2$$

$$\frac{\log N}{\log 2} = n - 1$$

$$n = \frac{\log 45}{\log 2} + 1$$

$$n = 6.49$$

$$n = \frac{\log N}{\log 2} + 1$$

** This works for both natural log and log base 10*

Practice Calculations

The ASTM Grain Size Number is 6.5. Approximately how many grains can be seen in a 1"x1" area at 85X mag.

Part 1: Find N (100x first)

$$N = 2^{n-1}$$

$$N = 2^{6.5-1}$$

$$N = 45 \text{ grains}$$

Part 2: Find N_{85}

$$N * 100^2 = N_{85} * 85^2$$

$$N_{85} = \frac{45 \text{ grains} * 100^2}{85^2}$$

$$N_{85} = 62.3 \text{ grains}$$

Is it better to have small or large grains?

“The yield strength of a metal increases as the grain size decreases. So, the formability of a component may decrease if the grain size gets too small.

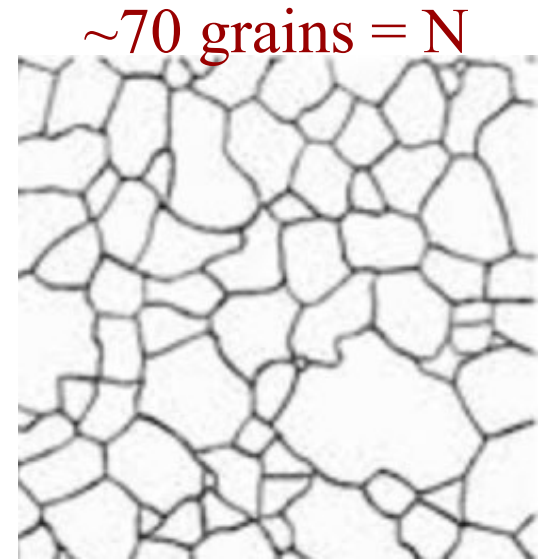
Structure changes properties

Larger grained metals have better formability, but the roughened surface that results from stretching metal with grain sizes that are too large is unacceptable for many applications. During a recrystallization anneal new grains form in the metal. The final grain size depends on the annealing temperature and annealing time.”

Processing changes structure

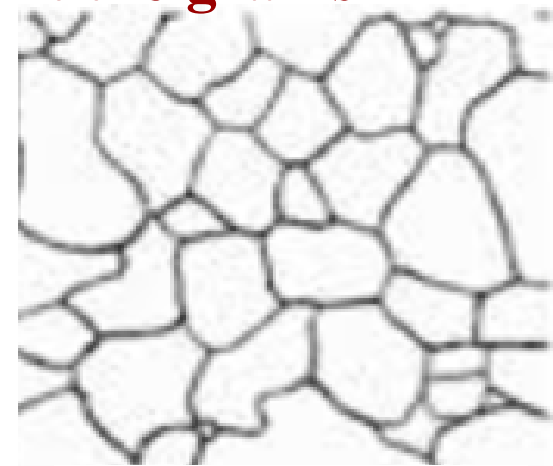
Let's Compare two processing temperatures

1. As shown in the figure below (1"x1" area under a 100X magnification microscope), a cold-rolled metal has been annealed at different temperatures for one hour. Calculate the ASTM Grain Size Number for the condition: Annealed at 550°C.



Step 1: count the grains

2. Now calculate the ASTM Grain Size Number for the condition: Annealed at 650°C:



$\sim 40 \text{ grains} = N$

Let's Compare two processing temperatures

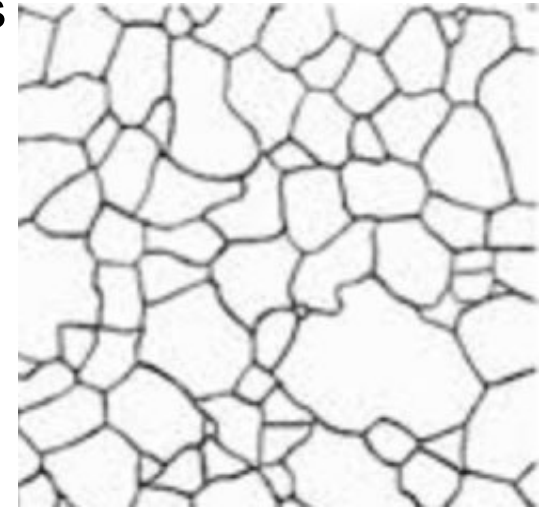
1. As shown in the figure below (1"x1" area under a 100X magnification microscope), a cold-rolled metal has been annealed at different temperatures for one hour. Calculate the ASTM Grain Size Number for the condition: Annealed at 550°C.

$$N = 2^{n-1} \longrightarrow n = \frac{\log N}{\log 2} + 1$$

$$n = 7.13$$

$$n = \frac{\log 70}{\log 2} + 1$$

~70 grains = N



2. Now calculate the ASTM Grain Size Number for the condition: Annealed at 650°C:

$$n = \frac{\log 40}{\log 2} + 1$$

$$n = 6.32$$

Step 1: count the grains



~40 grains³³ = N

Volume Defects (3D)

- gas pores, foreign inclusions, cracks ...
- **Example:**

